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1. Define sequence of real numbers.

Ans : A function whose domain is the set of natural numbers

i.e. $f: N \to R$ where R is set of real numbers

define a sequence of real numbers.

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2. Define monotone sequence.

Sol : A sequence $\{a_n\}$ is said to be monotonic if it is either monotonically increasing or monotonically decreasing.

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6. Give example of divergent series $\sum_{n=1}^{\infty} a_n$

such that $\lim_{n \to \infty} a_n = 0$

Sol : $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent,

but $\lim_{n \to \infty} \frac{1}{n} = 0$

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13. Define convergence of series of function $\sum_{n=1}^{\infty} u_n(x)$

Ans : The series $\sum_{n=1}^{\infty} u_n(x)$ is said to be convergent to f(x) if the sequence of partial sum $\{S_n(x)\}$ of $\sum_{n=1}^{\infty} u_n(x)$ is convergent to f(x).

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14.Define : Uniform convergent of sequence of functions.

Ans : The sequence $\{f_n\}_{n=1}^{\infty}$ of real valued functions on a set E , $\exists$, $\epsilon > 0$, $\exists$ positive integer N such that $|f_n(x) - f(x)| < \epsilon$, $\forall\, n \geq N$, $x \in E$ then $\{f_n\}_{n=1}^{\infty}$ is uniformly convergent to f on E

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15. Define pointwise convergence of a sequence of functions $\{f_n(x)\}_{n=1}^{\infty}$ on a set.

Ans : The sequence $\{f_n(x)\}_{n=1}^{\infty}$ is said to be point wise convergent to f(x) if for each $x \in E$ for given $\epsilon > 0$,

$\exists\ N \in I$ such that $|f_n(x) - f(x)| < \epsilon\ \forall\ n \geq N$

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16. Show that the sequence $\left\{\frac{nx}{n^2x^2+1}\right\}_{n=1}^{\infty}$ converges pointwise to zero.

Sol : $$\lim_{n\to\infty}\frac{nx}{n^2x^2+1} = \lim_{n\to\infty}\frac{nx}{n^2\left(x^2+\frac{1}{n^2}\right)}$$

$$= \lim_{n\to\infty}\frac{x}{n\left(x^2+\frac{1}{n^2}\right)}$$

$$= 0$$

$\therefore$ sequence $\left\{\frac{nx}{n^2x^2+1}\right\}_{n=1}^{\infty}$ converges pointwise to zero.

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17. Define when $\sum_{n=1}^{\infty} a_n$ converges absolutely ?

Ans : The series $\sum_{n=1}^{\infty} a_n$ converges absolutely

if $\sum_{n=1}^{\infty} |a_n|$ is convergent.

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18. When the series $\sum_{n=1}^{\infty} a_n$ converges conditionally ?

Ans : if $\sum_{n=1}^{\infty} a_n$ is convergent and

$\sum_{n=1}^{\infty} |a_n|$ is divergent then

$\sum_{n=1}^{\infty} a_n$ is said to be conditional convergent.

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19. State Root test for uniform convergence of $\sum_{n=1}^{\infty} a_n$

Sol : Let $\lim_{n\to\infty} \sup \sqrt[n]{|a_n|} = A$

if $A < 1$ then $\sum_{n=1}^{\infty} a_n$ is converges absolutely

if $A > 1$ then $\sum_{n=1}^{\infty} a_n$ is divergent

if $A = 1$ then test fails.

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21. State Weierstrass M-test for convergence (uniform) of functions.

Ans : The series $\sum_{k=1}^{\infty} u_k(x)$ *on* E is said to convergent uniformly if $\exists$ positive $M_1, M_2, M_3, \ldots$ with $\sum_{k=1}^{\infty} M_k < \infty$ such that $\sum_{k=1}^{\infty} u_k(x) \leq \sum_{k=1}^{\infty} M_k$

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22. State Cauchy's principle of uniform convergence of sequence of functions on E.

Ans : The sequence $\{f_n(x)\}_{n=1}^{\infty}$ is said to be uniformly convergent to f(x) if for each $x \in E$ for given $\epsilon > 0$, $\exists\ N \in I$ such that $|f_m(x) - f_n(x)| < \epsilon\ \forall\ m, n \geq N$

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23. Define uniformly convergence of a sequence of functions $\{f_n(x)\}_{n=1}^{\infty}$ on a set.

Ans : The sequence $\{f_n(x)\}_{n=1}^{\infty}$ is said to be uniformly convergent to f(x) if for each $x \in E$ for given $\epsilon > 0, \exists\ N \in I$ such that

$$|f_n(x) - f(x)| < \epsilon \ \forall\ n \geq N$$

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24. Define subsequence of a sequence of real numbers.

Ans : Consider f : N → I be sequence as $\{a_n\}_{n=1}^{\infty}$

and $g: N \to N$ be sequence as $\{n_i\}_{i=1}^{\infty}$.

Then $fog : N \to I$ defined as $\{a_{n_i}\}_{n=1}^{\infty}$ is

called a subsequence.

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25. Define : Rearrangement of series $\sum_{n=1}^{\infty} a_n$

Ans : The rearrangement of series $\sum_{n=1}^{\infty} a_n$ is a series $\sum_{n=1}^{\infty} b_n$ whose terms are exactly same as those of $\sum_{n=1}^{\infty} a_n$ but occurring in some different order .

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26. When $\sum_{n=1}^{\infty} u_n$ converges to F on E ?

Ans : $\sum_{n=1}^{\infty} u_n$ converges/ point wise to F on E

if $\lim_{n \to \infty} S_n$ converges to F on E

where $S_n = u_1 + u_2 + \ldots\ldots + u_n$

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27. When the series $\sum_{n=0}^{\infty} a_n$ is Abel summable to L ?

Ans : If $\lim_{x \to 1} f(x) = L$,

where $f(x) = \sum_{n=0}^{\infty} a_n x^n$ $\quad (0 \leq x < 1)$

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29. For the Fourier series expansion of $f(x)$ in the range $[-\pi, \pi]$ write down value of a_0

Ans : $$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)dx$$

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If a and b are two irrational numbers then a^b is irrational?

It is not true in general, because $\sqrt{2}$ is irrational, and so $(\sqrt{2})^{\sqrt{2}}$ is also irrational but, $((\sqrt{2})^{\sqrt{2}})^{\sqrt{2}} = 2$, which is rational.

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What is the difference between largest element and supremum?

Largest element always belongs to set, but supremum may or may not be an element of set.

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Every finite set is an ordered set?

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No, because {1, -1, i, -i} is a finite set but not an ordered set.

Correct statement is, every finite subset of all real numbers is an ordered set.

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32. Write down the Fourier series coefficient a_0 ,

where f(x) is odd function

Ans : $a_0 = 0$

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33. Write down the Fourier series coefficient a_n ,

where f(x) is odd function

Ans $a_n = 0$

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34. Write down the value of Fourier coefficient b_n in Fourier series expansion of even function f(x).

Ans : $b_n = 0$

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35. Find value of Fourier coefficient b_n, if f(x) is even.

Ans : $b_n = 0$

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Sum/product of two irrational numbers is irrational?

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It is not in true in general, because $\sqrt{2}$ and $-\sqrt{2}$ are two irrational numbers but their sum and product 0 and -2 respectively. Both are rational numbers.

www.thundershare.net

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Tell any set which is not complete ordered field.

Set of all rational numbers(Q) is not complete odder field, because it has a non empty bounded above subset, which has no supremum in Q.

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Tell any set which satisfies the property, intersection of its all lower bounds and upper bounds has only one element?

Every singleton {a} set in R, has lower bound $(-\infty, a]$, and upper bound $[a, \infty)$. Intrsetion of $(-\infty, a]$ and $[a, \infty)$ is {a}.

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Why set of all rational number is not a complete ordered field?

$S = \{x < \sqrt{2} | x \in Q\}$, this set is non empty bounded above in Q, but has no supremum in Q.

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41. For half range sine series, write down Fourier series for f(x).

Ans : $f(x) = \sum_1^\infty b_n \sin nx$ where $b_n = \frac{2}{\pi}\int_0^\pi f(x) \sin nx \, dx$

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How do you approximate a function using Taylor series representation?

What is the general term of a Taylor series?

tutorial.math.lamar.edu › calcII › tay...

Calculus II - Taylor Series - Pauls Online Math Notes

26-May-2020 · f(x)=∞∑n=0cn(x−a)n=c0+c1 (x−a) +c2(x−a)2+c3(x−a)3+c ... So, all the terms except the first are zero and we now know ...

www.math.smith.edu › ~rhaas

[PDF] Approximating functions by Taylor Polynomials.

First calculate the derivatives of sin x! You should find a pattern that makes this easy. derivative at x = 0 f (x). = sinx is 0 f (x). = cosx is 1 f (x).

math.libretexts.org › Calculus › 8.5:...

8.5: Taylor Polynomials and Taylor Series - Mathematics LibreTexts

20-Dec-2020 · Calculate the first four derivatives of f(x) at x=0. Then find the fourth order Taylor

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f_x and their first partial derivatives are differentiable at $\mathbf{U}_0$ and f, f_x, f_{x_n} and their first partial derivatives are differentiable at $\mathbf{X}_0$

Example Let (r, θ) be polar coordinates in the xy-plane; that is,

$$x = r\cos\theta, \quad y = r\sin\theta$$

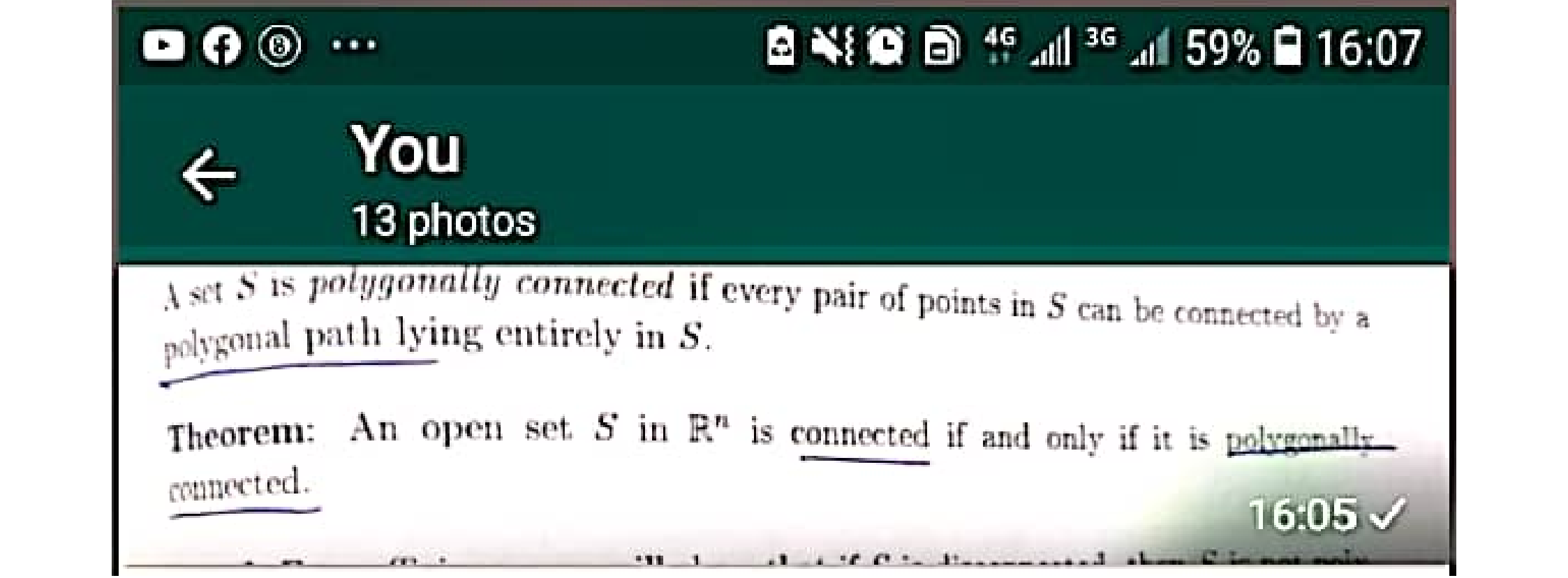
Suppose that $f = f(x, y)$ is differentiable on a set S, and let

$$h(r, \theta) = f(r\cos\theta, r\sin\theta).$$

We have

$$\frac{\partial h}{\partial r} \quad \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} \quad \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} \quad \frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y}$$

$\frac{\partial x}{\partial r} = \cos\theta$ $\frac{\partial y}{\partial r} = \sin\theta$ $\frac{\partial x}{\partial \theta} = r(-\sin\theta) = -r\sin\theta$ $\frac{\partial y}{\partial \theta} = r\cos\theta$ $2\pi\theta$



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Remark: The limit of a uniformly convergent series of integrable functions is integrable, and so term-by-term integration is permissible for such a series.

Theorem: Suppose that $\sum_{n=k}^{\infty} f_n$ converges uniformly to F on $S = [a, b]$. Assume that F and f_n, $n \geq k$, are integrable on $[a, b]$. Then

$$\int_a^b F(x)\,dx = \sum_{n=k}^{\infty} \int_a^b f_n(x)\,dx.$$

We say in this case that $\sum_{n=k}^{\infty} f_n$ can be integrated *term by term* over $[a, b]$.

Example: Consider the $\{F_n\}$ defined by

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$$\frac{1}{(1-x)^{k+1}} = \sum_{n=0}^{\infty} \binom{n+k}{k} x^n, \quad |x| < 1.$$

Example Show that the series

$$S(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \quad \text{and} \quad C(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

converges for all x.

Differentiating yields

$$S'(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^n}{(2n)!} = C(x)$$

and

$$C'(x) = \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n-1)!} = -\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = -S(x).$$

These results should not surprise you if you recall that

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The radius of convergence of each of these series is R. If

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad |x - x_0| < R,$$

then

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

1.10 The Taylor's Series

The only power series in $x - x_0$ that can possibly converge to f in such a hood is

$$\sum \frac{f^{(n)}(x_0)}$$

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Example: A similar argument shows that

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, \quad -\infty < x < \infty,$$

with uniform convergence on bounded sets.

Example: If $f(x) = e^x$, then $f^{(k)}(x) = e^x$ and $\|f^{(k)}\|_I = e^r$, $k \geq 0$, if $I = [-$

Since

$$\lim_{n \to \infty} \frac{r^n}{n!} e^r = 0,$$

we conclude that

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad -\infty < x < \infty,$$

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However, if we let (x, y) approach $(0,0)$ along the line $y = -x$, we see the functional values

$$f(x, -x) = -\frac{x^2}{2x^2} = -\frac{1}{2}$$

and conclude that the limit equals $-1/2$.

In fact, they are both incorrect. What we have shown is that

$$\lim_{x \to 0} f(x, x) = \frac{1}{2} \quad \text{and} \quad \lim_{x \to 0} f(x, -x) = -\frac{1}{2}.$$

Since $\lim_{x \to 0} f(x, x)$ and $\lim_{x \to 0} f(x, -x)$ must both equal $\lim_{(x,y) \to (0,0)} f(x, y)$

2.14 Vector Valued Functions

Suppose that $g_1, g_2, \ldots, g_n$ are real-valued functions defined on a subset T of $\mathbb{R}^m$.

We define the *vector-valued function* $\mathbf{G}$ on T by

$$\mathbf{G}(\mathbf{U}) = (g_1(\mathbf{U}), g_2(\mathbf{U}), \ldots, g_n(\mathbf{U})) \quad \mathbf{U} \in T.$$

Then $g_1, g_2, \ldots, g_n$ are the *component functions* of $\mathbf{G} = (g_1, g_2, \ldots, g_n)$. We

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f_{x_n} and the ... partial derivatives are differentiable at $\mathbf{U}_0$ and f, f_x, f_{x_2}, ..., f_{x_n} and their first partial derivatives are differentiable at $\mathbf{X}_0$.

Example: Let (r, θ) be polar coordinates in the xy-plane; that is,

$$x = r\cos\theta, \quad y = r\sin\theta$$

$\frac{\partial x}{\partial r} = \cos\theta$ $\frac{\partial y}{\partial r} = \sin\theta$ $\frac{\partial x}{\partial \theta} = r(-\sin\theta) = -r\sin\theta$ $\frac{\partial y}{\partial \theta} = r\cos\theta$

Suppose that $f = f(x, y)$ is differentiable on a set S, and let

$$h(r, \theta) = f(r\cos\theta, r\sin\theta).$$

We have

$$\frac{\partial h}{\partial r} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial r} = \cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y} \tag{2.85}$$

$$\frac{\partial h}{\partial \theta} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} = -r\sin\theta\frac{\partial f}{\partial x} + r\cos\theta\frac{\partial f}{\partial y}$$

where f_x and f_y are evaluated at $(x, y) = (r\cos\theta, r\sin\theta)$.

Example: Suppose that f_x and f_y just calculated are differentiable on an open set ... Differentiation (2.85) with respect to r yields

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Theorem: If $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are identically zero in an open region S of $\mathbb{R}^n$, then f is constant in S.

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Proof: We will show that if $\mathbf{X}_0$ and $\mathbf{X}$ are in S, then $f(\mathbf{X}) = f(\mathbf{X}_0)$

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$r \geq 1$ in $\mathbf{X} - \mathbf{X}_0$.

If $p(\mathbf{X}) \geq 0$ for all $\mathbf{X}$, we say that p is *positive semidefinite*; if $p(\mathbf{X}) \geq 0$ except when $\mathbf{X} = \mathbf{X}_0$, p is *positive definite*. Similarly, p is *negative semidefinite* if $p(\mathbf{X}) \leq 0$ or *negative definite* if $p(\mathbf{X}) < 0$ for all $\mathbf{X} \neq \mathbf{X}_0$. In all these cases, p is *semidefinite*. With p as in (2.102),

$$p(-\mathbf{X} + 2\mathbf{X}_0) = (-1)^r p(\mathbf{X}),$$

so p cannot be semidefinite if r is odd.

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Theorem: Suppose that f and g are locally integrable on $[a,b)$, $g(x) > 0$ and $f(x) \geq 0$ on some subinterval $[a_1, b)$ of $[a,b)$, and

$$\lim_{x \to b-} \frac{f(x)}{g(x)} = M. \tag{3.1}$$

- If $0 < M < \infty$, then $\int_a^b f(x)\,dx$ and $\int_a^b g(x)\,dx$ converge or diverge together.
- If $M = \infty$ and $\int_a^b g(x)\,dx = \infty$, then $\int_a^b f(x)\,dx = \infty$.
- If $M = 0$ and $\int_a^b g(x)\,dx < \infty$, then $\int_a^b f(x)\,dx < \infty$.

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3.2 Absolute integrability

We say that f is *absolutely integrable on* $[a,b)$ if f is locally integrable on $[a,b)$ and $\int_a^b |f(x)|\,dx < \infty$. In this case we also say that $\int_a^b$

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Theorem: If f is differentiable at $\mathbf{X}_0$, then f is continuous at $\mathbf{X}_0$.

Proof: From $L(\mathbf{X}) = m_1x_1 + m_2x_2 + \cdots + m_nx_n$, and Schwarz's ine

$$|L(\mathbf{X} - \mathbf{X}_0)| \leq M|\mathbf{X} - \mathbf{X}_0|,$$

where

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3.2 Absolute Integrability

We say that f is *absolutely integrable on* $[a, b)$ if f is locally integrable on $[a, b)$ and $\int_a^b |f(x)|\,dx < \infty$. In this case we also say that $\int_a^b f(x)\,dx$ converges absolutely or is *absolutely convergent*.

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3.9 Upper and Lower Integrals

If f is bounded on a rectangle R in $\mathbb{R}^n$ and $\mathbf{P} = \{R_1, R_2, \ldots, R_k\}$ is a partition of R.

Let

$$M_j = \sup_{\mathbf{X} \in R_j} f(\mathbf{X}), \quad m_j = \inf_{\mathbf{X} \in R_j} f(\mathbf{X}).$$

The *upper sum* of f over $\mathbf{P}$ is

$$S(\mathbf{P}) = \sum_{j=1}^{k} M_j V(R_j).$$

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The *upper integral of* f *over* R, denoted by

$$\overline{\int_R} f(\mathbf{X})\, d\mathbf{X},$$

is the infimum of all upper sums.

Upper and Lower Integrals: The *lower sum of* f *over* $\mathbf{P}$ is

$$s(\mathbf{P}) = \sum_{j=1}^{k} m_j V(R_j).$$

The *lower integral of* f *over* R, denoted by

$$\underline{\int_R} f(\mathbf{X})\, d\mathbf{X},$$

is the supremum of all lower sums.

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2.17.2 Differential

The linear function

$$L(\mathbf{X}) = f_{x_1}(\mathbf{X}_0)x_1 + f_{x_2}(\mathbf{X}_0)x_2 + \cdots + f_{x_n}(\mathbf{X}_0)x_n.$$

This function is called the *differential of* f *at* $\mathbf{X}_0$. We will denote it by $d_{\mathbf{X}_0}f$ and its value by $(d_{\mathbf{X}_0}f)(\mathbf{X})$.

Thus,

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2.17.4 Continuously Differentiable Function

We say that f is *continuously differentiable* on a subset S of $\mathbb{R}^n$ if S is contained i an open set on which $f_{x_1}, f_{x_2}, \ldots, f_{x_n}$ are continuous.

The above theorem implies that such a function is differentiable at each $\mathbf{X}_0$ in S

Example: If

$$f(x, y) = \frac{x^2 + y^2}{x - y},$$

then

$$f_x(x, y) = \frac{2x}{x - y} - \frac{x^2 + y^2}{(x - y)^2}$$

$$f_y(x, y) = \frac{2y}{x - y} + \frac{x^2 + y^2}{(x - y)^2}.$$

Since f_x and f_y are continuous on

$$S = \{(x, y) : x \neq y\}$$

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Show that (5)

$\frac{1}{2}[\log(1+x)]^2 = \frac{x^2}{2} - \frac{x^3}{3}\left(1+\frac{1}{2}\right) + \frac{x^4}{4}\left(1+\frac{1}{2}+\frac{1}{3}\right)+\cdots, \quad -1 < x \leq 1$

MTH 631

If X & Y are in $\mathbb{R}^n$ then prove that $|X \cdot Y| \leq |X||Y|$ with equality iff one of the vectors is a scalar multiple of the other (10)

What is the norm defined over a set of functions Further show that the sequence of functions $\{f_n(x)\}$ where $f_n(x) = \frac{x}{1+nx}$ converges pointwise on $[0, \infty]$ Find also the limit

(5)

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"mth631"

(10) (1) Def Riemann Sum and Riemann integral in R^n briefly.

(10) (2) Suppose $U_0 \in T$ is limit point of T, G is continuous at U_0 and f is continuous at $X_0 = G(U_0)$. Then show that fog is conti... at "U_0"

(5) (3) If $h(u,v) = f(g(u,v), -g(u,v))$, then prove that, $dh = (f_x - f_y)dg$.

(5) (4) Show that, $\int_1^\infty x^q \sin x\,dx$ diverges if $q > 0$

(5) (5) Determine weather $\int_1^\alpha \frac{\log x + \sin x}{\sqrt{x}}\,dx$ converge OR diverges.

(5) (6) Show that intervals are only region in R.

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MTH#631

Q No 1: (5 marks)

Suppose that if $\{f_n\}$ converges uniformly to f on $S = [a,b]$. Assume that if f_n and f are integrable on $[a,b]$ then prove that $\lim_{n\to\infty} \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx$

Q NO2: (5mark)

Prove that if $\{a_n x^n\}$ is bounded and $|x_1 - x_0| < r$ then $\sum a_n (x_1 - x_0)^n < r$ converges

Q NO 3: - (10 mark)

Derive the identity $2\sin x \cos x = \sin 2x$ from Maclaurin series for $\sin x$, $\cos x$ and $\sin 2x$.

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⑤ If $f_n(x) = A + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$ converges uniformly to $f(x)$ in $(-L, L)$. Show that $a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$ for $n = 1, 2, 3, \ldots$

⑤ Show that the series for which $S_n(x) = \frac{nx}{1+n^2x^2}$, $0 \le x \le 1$ cannot be differentiated at $x = 0$

⑩ Drive the identity $2\sinh x \cosh x = \sinh 2x$ from the Maclaurin's series for $\sinh x$, $\cosh x$, $\sinh 2x$.

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⑧ If $f_n(x) = A + \sum_{n=1}^{n} \left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right)$ converges uniformly to $f(x)$ in $(-L, L)$. Show that $a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\frac{n\pi x}{L}\,dx$ for $n = 1, 2, 3, \ldots$

⑨ Show that the series for which $S_n(x) = \frac{nx}{1+n^2x^2}$, $0 \le x \le 1$ cannot be differentiated term by term at $x = 0$

⑩ Drive the identity $2\sin x\cos x = \sin 2x$ from the Maclaurin's series for $\sin x$, $\cos x$, $\sin 2x$.

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots$$

$\sin x$

$f(x) = \sin x \Rightarrow 0$

$f'(x) = \cos x \Rightarrow 1$

$f''(x) = -\sin x \Rightarrow 0$

$f'''(x) = -\cos x \Rightarrow -1$

$\cos x$

$f(x) = \cos x$

$f'(x) = -\sin x$

$f''(x) = -\cos x$

$f'''(x) = \sin x$

$\sin 2x$

$f(x) = \sin 2x$

$f'(x) = 2\cos 2x$

$f''(x) = -4\sin 2x$

$f'''(x) = -8\cos 2x$

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Example: The functions

$$F_n(x) = \left(1 - \frac{nx}{n+1}\right)^{n/2}, \quad n \geq 1,$$

define a sequence on $D = (-\infty, 1]$.

1.2 Pointwise Convergence

Suppose that $\{F_n\}$ is a sequence of functions on D and the sequence of values $\{F_n(x)\}$ converges for each x in some subset S of D. Then we say that $\{F_n\}$ *converges pointwise on S to the limit function F*, defined by

$$F(x) = \lim_{n\to\infty} F_n(x), \quad x \in S.$$

Example: The sequence of functions defined by

$$\begin{cases} \infty, & x < 0 \end{cases}$$

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Moreover, if either g or h is bounded on S, then

$$\|g - h\|_S \geq |\|g\|_S - \|h\|_S|| .$$

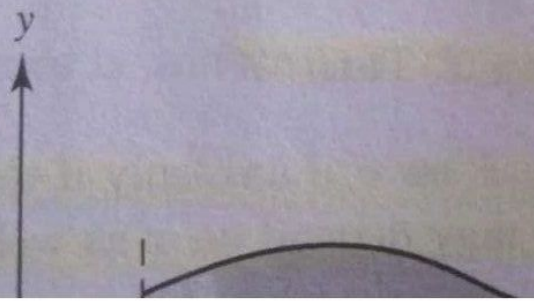
1.4 Uniform Convergence

A sequence $\{F_n\}$ of functions defined on a set S *converges uniformly to the limit function* F *on* S if

$$\lim_{n\to\infty} \|F_n - F\|_S = 0.$$

Thus, $\{F_n\}$ converges uniformly to F on S if for each $\varepsilon > 0$ there is an integer N such that

$$\|F_n - F\|_S < \varepsilon \quad \text{if} \quad n \geq N. \tag{1.1}$$



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lies in the strip

$$-\varepsilon \le y \le \varepsilon, \quad x \ge 0$$

Theorem: Let $\{F_n\}$ be defined on S. Then

1. $\{F_n\}$ converges pointwise to F on S if and only if there is, for each $\varepsilon > 0$ and $x \in S$, an integer N (which may depend on x as well as ε) such that

$$|F_n(x) - F(x)| < \varepsilon \quad \text{if} \quad n \ge N(\varepsilon, x).$$

2. $\{F_n\}$ converges uniformly to F on S if and only if there is for each $\varepsilon > 0$ an integer N (which depends only on ε and not on any particular x in S) such that

$$|F_n(x) - F(x)| < \varepsilon \quad \text{for all } x \text{ in } S \text{ if } n \ge N(\varepsilon).$$

Theorem: If $\{F_n\}$ converges uniformly to F on S, then $\{F_n\}$ converges pointwise to F on S.

The converse is false; that is, pointwise convergence does not imply uniform convergence.

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$$S_\rho = (-\infty, \rho] \cup [\rho, \infty)$$

for any $\rho > 0$, since

$$\|F_n - F\|_{S_\rho} = 0 \quad \text{if} \quad n > \frac{2}{\rho}.$$

How to show that a sequence of functions is not uniformly convergent?

Suppose that a sequence of function F_n is point wise convergent on the set S. Then the convergence of F_n is not uniform, if there exists an $\varepsilon > 0$ such that to each integer N there correspond and integer $n > N$ and a point $x_n \in S$ for which we have

$$|F_n(x_n) - F(x_n)| \geq \varepsilon.$$

MCQ

Example: If $F_n(x) = x^n$, $n \geq 1$, then $\{F_n\}$ converges pointwise on $S = [0, 1]$ to

$$F(x) = \begin{cases} 1, & x = 1, \\ 0, & 0 \leq x < 1. \end{cases}$$

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and $\lim_{n\to\infty} \rho^n = 0$. Another way to say the same thing: $\{F_n\}$ converges uniformly on every closed subset of $[0,1)$.

1.5 Cauchy's Uniform Convergence Criterion

Theorem: A sequence of functions $\{F_n\}$ converges uniformly on a set S if and only if for each $\varepsilon > 0$ there is an integer N such that

$$\|F_n - F_m\|_S < \varepsilon \quad \text{if} \quad n, m \geq N. \tag{1.3}$$

Proof: For necessity, suppose that $\{F_n\}$ converges uniformly to F on S.
Then, if $\varepsilon > 0$, there is an integer N such that

$$\|F_k - F\|_S < \frac{\varepsilon}{2} \quad \text{if} \quad k \geq N.$$

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Therefore, if $\|F_1 - F_0\|_S \frac{r^N}{1-r} < \varepsilon$,

then $\|F_n - F_m\|_S < \varepsilon$ if $n, m \geq N$.

1.6 Properties Preserved by Uniform Convergence

1.6.1 Continuity of the Limit Function at a Point

Theorem: If $\{F_n\}$ converges uniformly to F on S and each F_n is continuous at a point x_0 in S, then so is F. Similar statements hold for continuity from the right and left.

Proof: Suppose that each F_n is continuous at x_0. If $x \in S$ and $n \geq 1$, then

$$\begin{aligned} |F(x) - F(x_0)| &\leq |F(x) - F_n(x)| + |F_n(x) - F_n(x_0)| + |F_n(x_0) - F(x_0)| \\ &\leq |F_n(x) - F_n(x_0)| + 2\|F_n - F\|_S. \end{aligned}$$

(1.9)

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Example: Consider $F_n = \frac{1}{x^2+n^2}$ and [illegible]
by the series $\sum 1/n^2$ because

$$|F_n| < \frac{1}{n^2} =: M_n.$$

We know that $\sum 1/n^2 < \infty$.

1.8.3 Weierstrass M-test/dominated Convergence Test

Theorem The series $\sum f_n$ converges uniformly on S if

$$\|f_n\|_S \leq M_n, \quad n \geq k, \tag{1.20}$$

where $\sum M_n < \infty$.

Proof: From Cauchy's convergence criterion for series of constants, there is for each $\varepsilon > 0$ an integer N such that

$$M_n + M_{n+1} + \cdots + M_m < \varepsilon \quad \text{if} \quad m \geq n \geq N.$$

which, because of (1.20), implies [illegible]

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Solution: See the solution in video lecture.

Recall: If $\sum f_n$ converges uniformly on S, then $\lim_{n\to\infty} \|f_n\|_S = 0$. The series does not converge uniformly on $S = (-1/2, b)$ with $b < \infty$ or on $S = [a, \infty)$ with $a > -1/2$, because in these cases $\|f_n\|_S = 1$ for all n.

Absolute convergence: A series of functions $\sum f_n$ is said to *converge absolutely on* S if $\sum |f_n|$ converges pointwise on S, and converges *absolutely uniformly* on S if $\sum |f_n|$ converges uniformly on S.

Remarks:

- The condition of absolutely convergence (pointwise or uniform) is stronger than the usual convergence (pointwise or uniform).
- In our proof of Weierstrass's M-test, we actually proved that $\sum f_n$ converges absolutely uniformly on S.
- Show that if a series converges absolutely uniformly

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This is not surprising if you recognize that $D(x) = $.

Remark: Failure to verify that the given series converges at some point can lead to erroneous conclusions.

Example: For example, differentiating

$$\sum_{n=1}^{\infty} \cos \frac{x}{n} \tag{1.31}$$

MCQ

term by term.

We have

$$-\sum_{n=1}^{\infty} \frac{1}{n} \sin \frac{x}{n}.$$

Since

$$\left| \frac{1}{n} \sin \frac{x}{n} \right| \leq \frac{|x|}{n^2} \leq \frac{r}{n^2}, \quad |x| \leq r,$$

and $\sum 1/n^2 < \infty$. which converges uniformly on $[-r, r]$ for every r.

We cannot conclude from this that (1.31)

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where x_0 and $a_0, a_1, \ldots,$ are constants, is called a p

then power series becomes

$$\sum_{n=0}^{\infty} a_n x^n.$$

Theorem: The radius of convergence of $\sum a_n(x - x_0)^n$ is given by

$$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|$$

if the limit exists in the extended real number system.

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Theorem: For the power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$, define R in the extended real numbers by

$$\frac{1}{R} = \limsup_{n\to\infty} |a_n|^{1/n}. \tag{1.33}$$

Theorem: A power series

$$f(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n$$

R is continuous and differentiable in its interval

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0.

Then the power series converges

1. only for $x = x_0$ if $R = 0$;

2. for all x if $R = \infty$, and absolutely uniformly in every bounded set;

3. for x in $(x_0 - R, x_0 + R)$ if $0 < R < \infty$, and absolutely uniformly in every closed subset of this interval.

contrary

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Remark: The series diverges if $|x - x_0| > R$. No general statement can be made concerning convergence at the endpoints $x = x_0 + R$ and $x = x_0 - R$: the series may converge absolutely or conditionally at both, converge conditionally at one and diverge at the other, or diverge at both.

Theorem (Recall the following): Suppose that $\sum_{n=k}^{\infty} f_n$ converges uniformly to F on $S = [a, b]$. Assume that F and f_n, $n \geq k$, are integrable on $[a, b]$.

Then

$$\int_a^b F(x)\,dx = \sum_{n=k}^{\infty} \int_a^b f_n(x)\,dx.$$

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$$f(x) = \sum_{n=0} a_n(x - x_0)$$

then

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

1.10 The Taylor's Series

The only power series in $x - x_0$ that can possibly converge to f in such a neighborhood is

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n. \tag{1.43}$$

This is called the *Taylor series of f about x_0* (also, the *Maclaurin series* of f, if $x_0 = 0$). The mth partial sum of (1.43) is the Taylor polynomial

$$T_m(x) = \sum_{n=0}^{m} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n,$$

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The radius of convergence of each of these series is R. If

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad |x - x_0| < R,$$

then

$$a_n = \frac{f^{(n)}(x_0)}{n!}.$$

1.10 The Taylor's Series

The only power series in $x - x_0$ that can possibly converge to f in hood is

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$n=0$

where $R \geq \min\{R_1, R_2\}$.

Proof: See the video lectures.

Recall the following theorem:

Theorem: If $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ converge absolutely to sums A and B, th
the Cauchy product of $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ converges absolutely to AB.

Theorem: If f and g are given by power series

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n, \quad |x - x_0| < R_1,$$

$$g(x) = \sum_{n=0}^{\infty} b_n (x - x_0)^n, \quad |x - x_0| < R_2,$$

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$a_1 = 1, \quad a_3 = \frac{1}{6} + 2(1)$

erefore,

$$\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \cdots .$$

ample: To find the reciprocal of the power series

$$g(x) = 1 + e^x = 2 + \sum_{n=1}^{\infty} \frac{x^n}{n!},$$

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$$\frac{1}{1+e^x} = \frac{1}{2} - \frac{x}{4} + \frac{x^3}{48} + \cdots .$$

Example: To find the reciprocal of

$$g(x) = e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

we again let $h = 1$ in (1.51). If

$$(e^x)^{-1} = \sum_{n=0}^{\infty} a_n x^n,$$

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1.11. The Abel's Theorem

- If $\sum_{n=0}^{\infty} a_n R^n$ converges, then

$$\lim_{x \to (x_0+R)-} f(x) = \sum_{n=0}^{\infty} a_n R^n.$$

- If $\sum_{n=0}^{\infty} (-1)^n a_n R^n$ converges, then

$$\lim_{x \to (x_0-R)+} f(x) = \sum_{n=0}^{\infty} (-1)^n a_n R^n.$$

Proof: Let

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1.12 Pointwise and Uniform Bounded Functions

A sequence of functions $\{F_n\}$ on the set S is said to be pointwise bounded on S if the sequence of functions is bounded for every $x \in S$, that is, if there exists a finite valued function $\phi(x)$ defined on S such that

$$|F_n(x)| < \phi(x), \quad x \in S, n = 1, 2, 3, \ldots$$

We say that $\{F_n\}$ is uniformly bounded on S if there exist a number M such that

$$|F_n(x)| < M, \quad x \in S, n = 1, 2, 3, \ldots$$

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We say that $\{F_n\}$ is uniformly bounded on S if there exist a number M such that

$$|F_n(x)| < M, \quad x \in S, n = 1, 2, 3, ...$$

Remark: If $\{F_n\}$ is pointwise bounded on S and S_1 is countable subset of S, it is always possible to find a subsequence $\{F_{n_k}\}$ such that subsequence is convergent. However, even if $\{F_n\}$ is uniformly bounded sequence of continuous functions on a compact set S, there need not exist a subsequence which converges pointwise on S.

Example: Consider the sequence of functions

$$F_n(x) = \sin nx, \qquad x \in [0, 2\pi].$$

Suppose there exists a sequence $\{n_k\}$ such that $\{\sin n_k x\}$ converges, for every $x \in [0, 2\pi]$. Then we must have

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so no subsequence can converge uniformly on $[0, 1]$.

1.13 Equicontinuous Functions on a Set

A family of functions $\mathcal{F}$ defined on the set S is *equicontinuous* if for all $f \in \mathcal{F}$ and for each $\varepsilon > 0$ there is a $\delta > 0$ such that

$$|f(x_1) - f(x_2)| \leq \varepsilon \quad \text{if} \quad x_1, x_2 \in [a, b], \quad |x_1 - x_2| < \delta. \tag{1.58}$$

Remark: It is clear that every member of $\mathcal{F}$ is uniformly continuous.

Theorem: If $\{F_n\}$ is a pointwise bounded sequence of functions on a countable set S, then $\{F_n\}$ has a subsequence $\{F_{n_k}\}$ such that subsequence converges for all $x \in S$.

Proof: Let $\{x_i\}$, $i = 1, 2, 3, \ldots$ be the points of S arranged in a sequence.

Since $\{F_n(x_i)\}$ is bounded, there exists a subsequence

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By (3) property E is a subsequence of S_n, for $n = 1, 2, 3, \ldots$. The order in which the functions appear is the same in each sequence, i.e., if one function precedes another in S_1, they are in the same relation in every S_n, until one or the other is deleted.

Hence, when going from one row in the above array to the next below, functions may move to the left but never to the right. The (2) property of the sequence ensures that $\{F_{n,n}(x_i)\}$ converges as $n \to \infty$ for every $x \in S$.

Theorem: If K is a compact subset and if $\{F_n\}$ is a sequence of continuous functions defined on K and $\{F_n\}$ converges uniformly then $\{F_n\}$ is equicontinuous on K.

Proof: Since the sequence of functions $\{F_n\}$ is uniformly convergent, for every $\varepsilon > 0$, there is an integer N such that

$$\|F_n - F_N\|_K < \varepsilon, \qquad n > N.$$

We know that continuous functions on compact sets are uniformly continuous, there is a $\delta > 0$ such that

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$$
\begin{aligned}
|F_n(x) - F_n(y)| &\leq |F_n(x) - F_N(x)| + |F_N(x) - F_N(y)| \\
&\quad + |F_N(y) - F_n(y)| \\
&< 3\varepsilon.
\end{aligned}
$$

Theorem: If $\{F_n\}$ is a sequence of continuous functions defined on a compact S and if $\{F_n\}$ is a pointwise bounded and equicontinuous on S, then

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1.13. Equicontinuous Functions on a Set

1. $\{F_n\}$ is uniformly bounded on S,
2. $\{F_n\}$ contains a uniformly convergent subsequence.

Proof: Since $\{F_n\}$ is equicontinuous then by definition for every $\varepsilon > 0$,

$$|F_n(x) - F_n(y)| < \varepsilon, \qquad |x - y| < \delta.$$

From Analysis I, we know that S is compact then there are finitely ma
$p_1, p_2, \ldots p_r$ in S such that to every $x \in S$ corresponds at least one p_1
$|x - p_1| < \delta$.

Since $\{F_n\}$ is pointwise bounded, there exists $M_i < \infty$ such that

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there exists $M_i < \infty$ such that

$$|F_n(p_i)| < M_i, n \in \mathbb{N}.$$

If we take

$$M = \max\{M_1, ..., M_r\},$$

then $|F_n(x)| < M + \varepsilon$ for every $x \in S$. This proves the first part of the theorem.

Theorem: If $\{F_n\}$ is a sequence of continuous functions defined on a compact set S and if $\{F_n\}$ is a pointwise bounded and equicontinuous on S, then

1. $\{F_n\}$ contains a uniformly convergent subsequence.

Proof: Let E be a countable dense subset of S. Then from previous theorem we have a subsequence $\{F_{n_i}(x)\}$ such that the subsequence $\{F_{n_i}(x)\}$ converges for every $x \in E$.

Fix the notation $F_{n_i}(x) = g_i$, we shall prove that $\{g_i\}$ converges uniformly on S.

Let $\varepsilon > 0$, and choose δ as before. Let $V(x, \delta)$ be the set of all $y \in S$ such that $|x - y| < \delta$.

Since E is dense in S, and S is compact, there are finitely many points
in E such that

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1.14 The Stone-Weierstrass Theorem

Theorem: If f is continuous function on $[a, b]$, there exists a sequence of polynomials P_n such that

$$\lim_{x \to \infty} P_n(x) = f(x),$$

uniformly on $[a, b]$.

Proof: Without any loss of generality, we may assume that $[a, b] = [0, 1]$.

We may also assume that $f(0) = f(1) = 0$. As we can consider

$$g(x) = f(x) - f(0) - c[f(1) - f(0)], \qquad x \in [0, 1].$$

If g can be obtained as the limit of uniformly convergent sequence of polynomials, it is clear that the same is true for f since f

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Consider

$$\int_c^{c+\omega} f(x)dx = \int_c^0 f(x)dx + \int_0^\omega f(x)dx + \int_\omega^{c+\omega} f(x)dx$$

$$= -\int_0^c f(x)dx + \int_0^\omega f(x)dx + \int_0^c f(s)ds$$

$$\int_c^{c+\omega} f(x)dx = \int_0^\omega f(x)dx.$$

showing that the integral of a periodic function with period ω taken over an arbitrar

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2. $< (\mathbf{v}+\mathbf{u}), \mathbf{w} >=< \mathbf{v}, \mathbf{w} > + < \mathbf{u}, \mathbf{w} >$

3. $< c\mathbf{u}, \mathbf{v} >=< \mathbf{u}, c\mathbf{v} >= c < \mathbf{v}, \mathbf{u} >$

4. $< \mathbf{u}, \mathbf{u} > \geq 0$, and $< \mathbf{u}, \mathbf{u} >= 0$ if and only if $\mathbf{u} = 0$.

1.16 The space $\mathcal{E}$

Let us define the space $\mathcal{E}$ be the set of all real valued piecewise defined periodic function f on the interval $[-\pi, \pi]$.

Theorem: The space $\mathcal{E}$ is a linear space, that is, a vector space. Moreover, $\mathcal{E}$ an inner product space with respect to the inner product

$$< f, g >= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)g(x)dx.$$

The trigonometric functions: The set of functions

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1.16.1 Fourier Series of Even and Odd Functions

Even and odd functions possess certain simple but useful properties:

- The product of two even (or odd) functions is an even function.
- The sum of two even (or odd) functions is an even (or odd) function.
- The product of an even and an odd function is an odd function.
- For a Riemann integrable function f defined on $[-c, c]$ $(c > 0)$, it is evident that

$$\int_{-c}^{c} f(x)dx = 2\int_{0}^{c} f(x)dx, \quad \text{if } f \text{ is even}$$

$$\int_{-c}^{c} f(x)dx = 0, \quad \text{if } f \text{ is odd}$$

Fourier series of even function: Suppose that $f(x)$ is a periodic function of

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$$|x| = \frac{\pi}{2} - \frac{4}{\pi}\sum_{k=0}^{\infty}\frac{\cos(2k+1)x}{(2k+1)^2}.$$

Remark: Note that the Fourier series here converges uniformly to $|x|$ on $[-\pi, \pi]$ but not on the whole interval $(-\infty, \infty)$, and so outside the interval $(-\infty, \infty)$, $f(x)$ is determined by the periodicity condition $f(x) = f(x + 2\pi)$.

we can make use of this series to find the values of some numerical series. For instant $x = 0$ gives

$$\frac{\pi^2}{8} = \sum_{k=1}^{\infty}\frac{1}{(2k+1)^2}.$$

Some natural questions arise:

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In 1876, Paul du Bois-Reymond constructed a continuous function $f : [-\pi, \pi] \to \mathbb{R}$ whose Fourier series failed to converge to f at each point in a dense subset of $[-\pi, \pi]$.

Indeed, the following are true statements

- There exists a continuous function whose Fourier series diverges at a point.
- There exists a continuous function whose Fourier series converges everywhere on $[-\pi, \pi]$, but not uniformly.
- There exists a continuous function whose Fourier series diverges for points in some set S and converges on $(-\pi, \pi) \setminus S$.

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Thus, the Fourier series of f converges to f(x) at the point x where it is continuous.

At the point of discontinuity x, the Fourier series of f assumes the mean of the one-sided limits of f.

Corollary: If $f : [-\pi, \pi] \to \mathbb{R}$ is continuous, and if $f(-\pi) = f(\pi)$, $f'(x)$ exists and is piecewise continuous on $[-\pi, \pi]$, then the Fourier series of f converges to $f(x)$ at every point $x \in [-\pi, \pi]$.

Theorem: Suppose that $f : [-\pi, \pi] \to \mathbb{R}$ is piecewise continuous on $[-\pi, \pi]$ and piecewise monotone,that is, there exists a partition $P = x_0, x_1, ..., x_n$ of $[-\pi, \pi]$ such that the restriction $f|_{[x_{k-1}, x_k]}, k = 1, 2..., n$, is either increasing or decreasing.

Let $f(x)$ be defined for other values of x by the periodicity condition $f(x) = f(x + 2\pi)$. Then the Fourier series of f on $[-\pi, \pi]$ converges to

- $f(x)$ if f is continuous at $x \in (-\pi, \pi)$.
- $(f(x+) + f(x-))/2$ if f is discontinuous at x.
- $(f(\pi+) + f((-\pi)-))/2$ if f is discontinuous at $x = \pm\pi$.

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- f is odd function, hence $a_n = 0$.
- $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} x \sin nx dx = \frac{2}{\pi}\int_0^{\pi} x \sin nx dx = \frac{2(-1)^{n-1}}{n}$.

$$x \cong 2\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \sin kx.$$

Remarks: Note that the Fourier series does not necessarily agree with $f(x) = x$ at every point in $[-\pi, \pi]$.

The Fourier series vanishes at both endpoints $x = \pm\pi$, whereas the function does not vanish at either endpoint.

However, the Dirichlet's theorem states that series converges to $f(x)$ at every interior point of $(-\pi, \pi)$.

For example at $x = \pi/2$ the symbol $\cong$ could be replaced by $=$ and so

$$\frac{\pi}{2} = 2\left(1 - \frac{0}{2} + \frac{(-1)}{3} - \frac{0}{4} + \frac{1}{5} + \ldots\right).$$

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which reduces to

$$\frac{\pi \coth \pi - 1}{2} = \sum_{n=1}^{\infty} \frac{1}{(1+n^2)}.$$

1.17 Fourier Series for Arbitrary Periodic Function

Suppose that f is a $2L$-periodic and Riemann integrable function. The function $f(at)$ has period $2L/a$.

In particular, $f((L/\pi)t)$ is 2π-periodic, and so the Fourier series expansion has the following in terms of the variable t:

$$f(\frac{L}{\pi}t) \cong \frac{a_0}{2} + \sum_{n=1}^{\infty} \Big(a_n \cos nt + b_n \sin nt \Big), \qquad t \in [-\pi, \pi],$$

where

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(\frac{L}{\pi}t) \cos nt dt = \frac{1}{L} \int_{-L}^{L} f(x) \cos(\frac{k\pi}{L}x) dx.$$

$$f(\frac{L}{\pi}t) \cong \frac{a_0}{2} + \sum_{n=1}^{\infty} \Big(a_n \cos nt + b_n \sin nt \Big), \qquad t \in [$$

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Changing the variable t, by setting $t = (\pi/L)x$.

Theorem: Let f be a periodic function with period $2L$. Then the Fourier expansion of f is given by

$$f(x) \cong \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos(\frac{n\pi}{L}x) + b_n \sin(\frac{n\pi}{L}x)\right), \qquad x \in [-L, L],$$

where

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos(\frac{n\pi}{L}x)dx,$$

and

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin(\frac{n\pi}{L}x)dx.$$

Remark: The interval of integration in the last formulas for the Fourier coefficients can be replaced with the interval $[c, c+2L]$, where c is any real number; we usually let $c = -L$. Notice that

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$$b_n = \frac{1}{L}\int_{-L} f(x)\sin(\frac{k\pi}{L}x)dx.$$

we obtain

$$a_0 = \frac{1}{2}, \qquad a_n = 0.$$

$$b_n = \frac{1+(-1)^{n-1}}{n\pi}, \qquad n \geq 1.$$

Example: Consider the function $f(x) = |\sin x|$. The function is defined for all x and the function has period π.

Clearly, f represents a continuous, piecewise smooth, even function of period π, and therefore it is everywhere equal to its Fourier series, consisting of cosine terms only.

We have $c = 0$, and $L = \pi/2$, then we have

$$\begin{aligned} a_k &= \frac{2}{\pi}\int_0^{\pi} f(x)\cos(2kx)dx \\ &= \frac{2}{\pi}\int_0^{\pi} \sin x \cos(2kx)dx \end{aligned}$$

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Thus, the Fourier series expansion of $|\sin x|$ is

$$|\sin x| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos 2kx}{4k^2-1}, \qquad x \in [-\pi, \pi].$$

1.18 Best Approximation Theorem

Theorem: Let $\Phi = \phi_1, ..., \phi_n$ be an orthonormal set of functions in the inner product space $\mathcal{E}$, and let c_k be the Fourier coefficients of f relative to ϕ_k:

$$c_k = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\phi_k(x)dx := < f, \phi_k > .$$

If $T_n(x)$ is an arbitrary Fourier polynomial relative to ϕ_k, that is, $T_n(x) = \sum_{k=1}^{n} d_k\phi_k(x)$ for some constants $d_1, ..., d_n$, then we have

$$\left\| f - \sum c_k\phi_k(x) \right\|^2 \leq \|f - T_n\|^2,$$

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In particular, $|\mathbf{X}|$ is the distance [illegible]

unit vector.

Example: The lengths of the vectors

$$\mathbf{X} = (1, -2, 6, 5) \quad \text{and} \quad \mathbf{Y} = (3, -5, 4, \tfrac{1}{2})$$

are

$$|\mathbf{X}| = (1^2 + (-2)^2 + 6^2 + 5^2)^{1/2} = \sqrt{66}$$

and

$$|\mathbf{Y}| = (3^2 + (-5)^2 + 4^2 + (\tfrac{1}{2})^2)^{1/2} = \frac{\sqrt{201}}{2}.$$

The distance between $\mathbf{X}$ and $\mathbf{Y}$ is

$$|\mathbf{X} - \mathbf{Y}| = ((1-3)^2 + (-2+5)^2 + (6-4)^2 + (5-\tfrac{1}{2})^2)^{1/2} = \frac{\sqrt{149}}{2}.$$

The *inner product* $\mathbf{X} \cdot \mathbf{Y}$ of $\mathbf{X} = (x_1, x_2, \ldots, x_n)$ and $\mathbf{Y} = (y_1, y_2, \ldots, y_n)$ is

$\mathbf{X} \cdot \mathbf{Y}$

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$$\mathbf{X} \cdot \mathbf{Y} = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n.$$

2.2 Schwarz's Inequality

Lemma: If $\mathbf{X}$ and $\mathbf{Y}$ are any two vectors in $\mathbb{R}^n$, then

$$|\mathbf{X} \cdot \mathbf{Y}| \leq |\mathbf{X}|\,|\mathbf{Y}|, \tag{2.3}$$

with equality if and only if one of the vectors is a scalar multiple of the other.

Proof: Suppose that $\mathbf{Y} \neq \mathbf{0}$ and t is any real number. Then

$$\begin{aligned} 0 &\leq \sum_{i=1}^{n} (x_i - t y_i)^2 \\ &= \sum_{i=1}^{n} x_i^2 - 2t \textstyle\sum_{i=1}^{n} x_i y_i + t^2 \sum_{i=1}^{n} y_i^2 \\ &= |\mathbf{X}|^2 - 2(\mathbf{X} \cdot \mathbf{Y})t + t^2 |\mathbf{Y}|^2. \end{aligned} \tag{2.4}$$

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with equality if and only if one of the vectors is a nonnegative multiple of the ot

Proof: By definition,

$$\begin{aligned}
|\mathbf{X}+\mathbf{Y}|^2 &= \textstyle\sum_{i=1}^{n}(x_i+y_i)^2 \\
&= \textstyle\sum_{i=1}^{n} x_i^2 + 2\sum_{i=1}^{n} x_i y_i + \sum_{i=1}^{n} y_i^2 \\
&= |\mathbf{X}|^2 + 2(\mathbf{X}\cdot\mathbf{Y}) + |\mathbf{Y}|^2 \\
&\leq |\mathbf{X}|^2 + 2|\mathbf{X}|\,|\mathbf{Y}| + |\mathbf{Y}|^2 \quad \text{(by Schwarz's inequality)} \\
&= (|\mathbf{X}| + |\mathbf{Y}|)^2.
\end{aligned}$$

Hence,

$$|\mathbf{X}+\mathbf{Y}|^2 \leq (|\mathbf{X}| + |\mathbf{Y}|)^2.$$

Taking square roots yields (2.7).

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$$\lim_{r\to\infty} \mathbf{X}_r = \overline{\mathbf{X}}.$$

Theorem: Let

$$\overline{\mathbf{X}} = (\overline{x}_1, \overline{x}_2, \ldots, \overline{x}_n) \quad \text{and} \quad \mathbf{X}_r = (x_{1r}, x_{2r}, \ldots, x_{nr}), \quad r \geq 1.$$

Then $\lim_{r\to\infty} \mathbf{X}_r = \overline{\mathbf{X}}$ if and only if

$$\lim_{r\to\infty} x_{ir} = \overline{x}_i, \quad 1 \leq i \leq n;$$

that is, a sequence $\{\mathbf{X}_r\}$ of points in $\mathbb{R}^n$ converges to a limit $\overline{\mathbf{X}}$ if and only if the sequences of components of $\{\mathbf{X}_r\}$ converge to the respective components of $\overline{\mathbf{X}}$.

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ines
$\neq 1$

... radius r about $\mathbf{X}_0$,

$$\overline{S}_r(\mathbf{X}_0) = \{\mathbf{X} : |\mathbf{X} - \mathbf{X}_0| \leq r\}.$$

Remark: Open and closed n-balls are generalizations to $\mathbb{R}^n$ of open and closed intervals.

Lemma: If $\mathbf{X}_1$ and $\mathbf{X}_2$ are in $S_r(\mathbf{X}_0)$ for some $r > 0$, then so is every point on the line segment from $\mathbf{X}_1$ to $\mathbf{X}_2$.

Proof: The line segment is given by

$$\mathbf{X} = t\mathbf{X}_2 + (1-t)\mathbf{X}_1, \quad 0 < t < 1.$$

Suppose that $r > 0$. If

$$|\mathbf{X}_1 - \mathbf{X}_0| < r, \quad |\mathbf{X}_2 - \mathbf{X}_0| < r,$$

and $0 < t < 1$, then

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2.5. Principle of nested sets

Theorem (Cauchy's Convergence Criterion): A sequence $\{\mathbf{X}_r\}$ converges if and only if for each $\varepsilon > 0$ there is an integer K such that

$$|\mathbf{X}_r - \mathbf{X}_s| < \varepsilon \quad \text{if} \quad r, s \geq K.$$

Diameter of a Set: If S is a nonempty subset of $\mathbb{R}^n$, then

$$d(S) = \sup\{|\mathbf{X} - \mathbf{Y}| : \mathbf{X}, \mathbf{Y} \in S\}$$

is the *diameter* of S.

If $d(S) < \infty$, S is *bounded*; if $d(S) = \infty$, S is *unbounded*.

2.5 Principle of nested sets

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Long Question:— Mth 631 10:30

Find Radius of convergence of (5)
Power Series $\sum_{n=1}^{\infty} 2^n p^n (x+1)^n$

Find Radius of convergence of
$L=2$
(i) $\sum_{K=0}^{\infty} aK x^K$ (ii) $\sum_{K=0}^{\infty} Pa_K x^K$

Expand $f(x) = x^2 \quad 0 < x < 2\pi$
in a Foleries Series having Period 2π

Mcq کے لیے تمام Series کو اوپر دی گئی طرح یاد کریں
اور Taylor's Series کی form کو یاد کیا ہے۔ اور
Mn میں Weierstrass M-test استعمال ہوتا ہے etc.

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Date:

Mth-631

Q.1

Check the following series functions

$$\sum \frac{1}{x^2+n^2} \quad \text{and} \quad \sum \frac{\sin nx}{n^2}$$

are uniform convergence or not?

Ans:-

by dominated series we have

$$\frac{1}{x^2+n^2} \leq \frac{1}{n^2} \quad , \quad \left|\frac{\sin nx}{n^2}\right| \leq \frac{1}{n^2}$$

Taking $M_n = \frac{1}{n^2}$ and recall it

$$\sum M_n = \sum \frac{1}{n^2} < \infty$$ **M.M**

by Weierstrass M-test, we can conclude

$$\sum \frac{1}{x^2+n^2} \quad \text{and} \quad \sum \frac{\sin nx}{n^2}$$

converge uniformly on $(-\infty, \infty)$.

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Q.2

Check the uniform convergence of the series

$$\sum f_n(x) = \sum \left(\frac{x}{1+x}\right)^n$$

Soln:-

The series is uniformly convergence on set S such that

$$\left|\frac{x}{1+x}\right| \leq \gamma < 1 \quad , \quad x \in S \qquad \longrightarrow (i)$$

M-M

For set S we have $\|f_n\|_S \leq \gamma^n$

by weierstrass M-test $\sum M_n = \sum \gamma^n < \infty$

(i) will be

$$\frac{-\gamma}{1+\gamma} \leq x \leq \frac{\gamma}{1-\gamma} \quad , \quad x \in S$$

put $\gamma = 1$ then

$$\frac{-1}{2} \leq x \leq \infty$$

This mean the series converge uniformly on any compact subset of $\left(-\frac{1}{2}, \infty\right)$.

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Mth-6031

Q.3

Define Absolute convergence?

Ans:-

A series of function Σf_n is said to be Absolute convergence on S if $\Sigma |f_n|$ converges pointwise on S and converges absolutely uniformly on S if $\Sigma |f_n|$ converges uniformly on S.

Q.4 M.M

Evaluate $\sum_{n=1}^{\infty} \int_0^1 x(e^x - 1)e^{-nx}\,dx$

Sol:-

Part-1

Given

$$\sum_{n=1}^{\infty} \int_0^1 x(e^x-1)e^{-nx}\,dx$$

converge it to $F(x) = x$.

by the theorem of interchange of sum and integral sign, we have

$$\sum_{n=1}^{\infty} \int_0^1 x(e^x-1)e^{-nx}\,dx$$

$$= \int_0^1 \sum_{n=1}^{\infty} x(e^x-1)e^{-nx}\,dx$$

$$= \int_0^1 x\,dx$$

Part-2

$$= \left[\frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{2} - 0$$

$$= \frac{1}{2}$$

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Q.45

If $\sum_{n=K}^{\infty} f_n$ converges uniformly to F on S and each f_n is continuous at a point x_0 in S, then so is F. similar statements hold for continuity from the right and left?

Ans.:-

Since it is proved in Q.2 that

$$F(x) = \sum_{n=0}^{\infty} \left(\frac{x}{1+x}\right)^n$$ M.M

is uniformly converge on every compact subset of $(\frac{-1}{2}, \infty)$.

As term of series is continuous on every such subset so it implies that F is also.

In fact we can state a stronger result, F is continuous on $(-\frac{1}{2}, \infty)$. since every point lies in $(-\frac{1}{2}, \infty)$ and it also lies in compact subinterval of $(-\frac{1}{2}, \infty)$.

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Q.6

The series $E(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$ is converges uniformly on interval $[-r, r]$

soln:-

The series converges uniformly on every interval $[-r, r]$

by ~~weier~~ weierstrass M-test

$$\frac{|x|^n}{n!} \leq \frac{r^n}{n!}$$

M-M

$$|x|^n \leq r^n \implies |x| \leq r$$

So, $\sum \frac{r^n}{n!} \leq \infty$

by Diff.

$$\sum_{n=1}^{\infty} \frac{x^n}{n!} = \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!}$$

Therefore, the differential series is also uniformly converges on $[-r, r]$.

Q.7

Power series :-

An infinite series of the form $\sum_{n=0}^{\infty} a_n (x - x_0)^n$ where x_0 and $a_0, a_1, \cdots a_n$ are constant is called a power series in $x - x_0$ if $x_0 = 0$ then power series will be

$$\sum_{n=0}^{\infty} a_n x^n$$

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Mth-631

Q.8

Lemma: If $f(x)$ is a periodic function with period ω, then $\int_c^{c+\omega} f(x)dx = \int_0^{\omega} f(x)dx$ whenever f is integrable on $[0, \omega]$?

Solu-

Given

$$\int_c^{c+\omega} f(x)dx = \int_0^{\omega} f(x)dx$$

M.M

L.H.S

$$\int_c^{c+\omega} f(x)dx = \int_c^{0} f(x)dx + \int_0^{\omega} f(x)dx + \int_{\omega}^{c+\omega} f(x)dx$$

$$= -\int_0^{c} f(x)dx + \int_0^{\omega} f(x)dx + \int_0^{c} f(x)dx$$

$$= \int_0^{\omega} f(x)dx$$

It show that the integral of a periodic functi with period ω taken over an arbitrary interval of length ω always has the same value.

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Mth-631

Q.9

Write the properties of fourier series of Even and odd function:-

Ans:-

i) The product of two even or odd functions is an even function.

ii) The sum of two even or odd functions is an even or odd function.

iii) The product of even and odd function is odd function.

M.M

Q.11

The length of vectors is $X=(1,-2,6,5)$ and $Y=(3,-5,4,\frac{1}{2})$, Find the distance b/w X and Y and Find inner product.

Soln:-

Distance

$$|X-X| = \sqrt{(1-3)^2+(-2+5)^2+(6-4)^2+(5-\frac{1}{2})^2}$$

$$= \sqrt{4+9+4+\frac{81}{4}} = \frac{\sqrt{149}}{2}$$

inner product

$$X \cdot Y = x_1y_1 + x_2y_2 + x_3y_3 + x_4y_4$$

$$= (1)(3)+(-2)(-5)+(6)(4)+(5)(\tfrac{1}{2})$$

$$= 3+10+24+\frac{5}{2} = \frac{79}{2} = 39\frac{1}{2}$$

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Q.10

Prove the function $f(x) = |\sin x|$ is defined for all x and has period π.

Sol:-

Clearly, f represent a continuous and even function of period π, therefore it is everywhere equal to its fourier series, containing Cosine terms only.

here we have $c=0$ and $L=\frac{\pi}{2}$, then

$$a_k = \frac{2}{\pi}\int_0^{\pi} f(x)\cos(2kx)\,dx$$

M.M

$$= \frac{2}{\pi}\int_0^{\pi} \sin x \cos(2kx)\,dx$$

$$= \frac{1}{\pi}\int_0^{\pi} \left[\sin(1+2k)x - \sin(2k-1)x\right]dx$$

$$= \frac{1}{\pi}\left(-\frac{\cos(1+2k)x}{2k+1} + \frac{\cos(2k-1)x}{2k-1}\right)\Bigg|_0^{\pi}$$

$$= -\frac{1}{\pi}\left(\frac{(-1)^{2k+1}-1}{2k+1} - \frac{(-1)^{2k-1}-1}{2k-1}\right)$$

$$= -\frac{4}{\pi(4k^2-1)}$$

Thus fourier series of $|\sin x|$ is

$$|\sin x| = \frac{2}{\pi} - \frac{4}{\pi}\sum_{k=1}^{\infty}\frac{\cos 2kx}{4k^2-1}, \quad x \in [-\pi, \pi]$$

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A PROOF OF $\sin 2x = 2\sin x \cos x$

ÁNGEL PLAZA

Here we show that a trigonometric identity as $\sin 2x = 2\sin x \cos x$, may be proved ultimately by combinatorial arguments. The proof only uses the Taylor expansions of $\sin x$, $\cos x$, and the Cauchy product of two power series. That is,

$$\left(\sum_{n=0}^{\infty} a_n x^n\right) \cdot \left(\sum_{n=0}^{\infty} b_n x^n\right) = \sum_{n=0}^{\infty} c_n x^n, \text{ where } c_n = \sum_{k=0}^{n} a_k b_{n-k}$$

Thus if we write

$$\begin{aligned} \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \end{aligned}$$

by the Cauchy product, $\cos x \sin x$ can be written:

$$\cos x \sin x = \sum_{n=0}^{\infty} c_n x^n, \text{ where } c_n = \left\{ \begin{array}{cc} \displaystyle\sum_{k=0}^{m} \frac{(-1)^k (-1)^{m-k}}{(2k+1)!(2m-2k)!} & \text{if } \quad n = 2m+1 \\ 0 & \text{if } \quad n = 2m \end{array} \right\}$$

Since

$$\frac{1}{2}\sin 2x = x - \frac{2^2 x^3}{3!} + \frac{2^4 x^5}{5!} - \frac{2^6 x^7}{7!} + \ldots = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n}}{(2n+1)!} x^{2n+1}$$

the only thing we have to prove is the identity

$$\sum_{k=0}^{m} \frac{1}{(2k+1)!(2m-2k)!} = \frac{2^{2m}}{(2m+1)!}$$

Or, equivalently,

$$\begin{aligned} \sum_{k=0}^{m} \frac{(2m+1)!}{(2k+1)!(2m-2k)!} &= 2^{2m} \\ \sum_{k=0}^{m} \binom{2m+1}{2k+1} &= 2^{2m} \end{aligned}$$

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The last equation is a consequence of the facts:

$$(1+1)^{2m+1} = \sum_{k=0}^{2m+1} \binom{2m+1}{k} = 2^{2m+1} \tag{1}$$

$$(1-1)^{2m+1} = \sum_{k=0}^{2m+1} (-1)^k \binom{2m+1}{k} = 0 \tag{2}$$

By Eq. (2), $\sum_{k=0}^{m} \binom{2m+1}{2k+1} = \sum_{k=0}^{m} \binom{2m+1}{2k}$, and using Eq. (1), the last sums are both equal to 2^{2m} and the proof is done.

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Lecture 25:

(*) → **Uniform Continuity:**

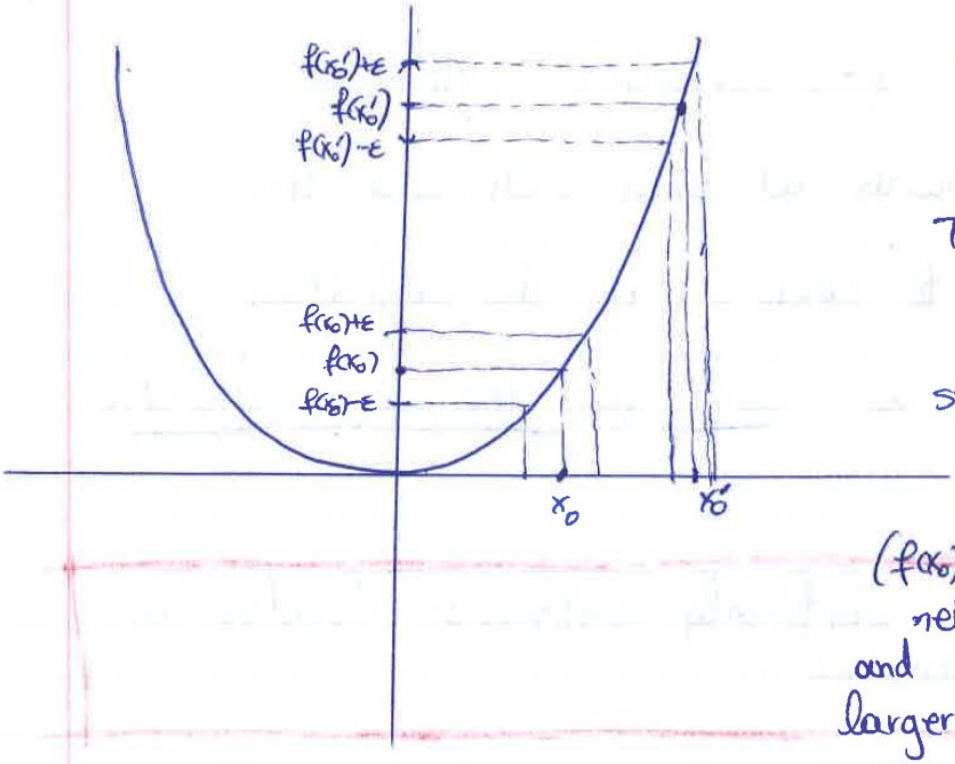
Let $f: A \subseteq \mathbb{R} \longrightarrow \mathbb{R}$. Let $x_0 \in A$.

Let us remember what it means that f is continuous at x_0: it means that, $\forall \varepsilon > 0$, there exists $\delta = \delta(x_0, \varepsilon) > 0$ (i.e., $\delta > 0$ potentially depending on x_0 and ε), such that: if $x \in A$ with $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.

There are of course examples of functions (such as $f(x) = x \ \forall x \in \mathbb{R}$, or $f(x) = c \ \forall x \in \mathbb{R}$, see old notes on continuity) where $\forall \varepsilon > 0$ we can pick a $\delta = \delta(\varepsilon) > 0$, i.e. a δ that only depends on ε, not on x_0, to satisfy the definition of continuity at x_0.

However, other functions (such as $f(x) = x^2 \ \forall x \in \mathbb{R}$) don't satisfy this:

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The larger x_0 is, the faster x_0^2 grows.

Thus, for fixed $\varepsilon>0$,

while, when x_0 is small, quite a large neighbourhood of x_0 is sent inside $(f(x_0)-\varepsilon, f(x_0)+\varepsilon)$, this neighbourhood shrinks more and more as x_0 gets larger and larger.

So, for such a function, when we test for continuity at x_0, the $\delta>0$ that corresponds to each $\varepsilon>0$ has to depend on x_0, not just on ε.

→ **Def: (Uniform Continuity):**

Let $f: A \subseteq \mathbb{R} \longrightarrow \mathbb{R}$. We say that f is uniformly continuous if:

for all $\varepsilon>0$, there exists $\delta=\delta(\varepsilon)>0$,

s.t. if $x, y \in A$, with $|x-y|<\delta$,

then $|f(x)-f(y)|<\varepsilon$.

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In other words, f is uniformly continuous if it is continuous at each $x_0 \in A$, and the δ corresponding to each ε in the definition of continuity at x_0 can be the same $\forall x_0 \in A$.

Observation: Each uniformly continuous function is continuous.

Proof: Let $f: A \subset \mathbb{R} \longrightarrow \mathbb{R}$ be uniformly continuous.

Let $x_0 \in A$. Let $\varepsilon > 0$.

Since f is uniformly continuous, for this $\varepsilon > 0$ there exists $\delta (= \delta(\varepsilon)) > 0$, such that:

If $x, y \in A$, and $|x-y| < \delta$, then $|f(x) - f(y)| < \varepsilon$.

In particular, if $x \in A$ and $|x - x_0| < \delta$, then $|f(x) - f(x_0)| < \varepsilon$.

Since $\varepsilon > 0$ was arbitrary, f is continuous at x_0.

Since $x_0 \in A$ was arbitrary, f is continuous on A. ■

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Lecture 26

Examples:

- $f(x)=x, \forall x\in\mathbb{R}$: uniformly continuous.
- $f(x)=x^2, \forall x\in\mathbb{R}$: not uniformly continuous.
- $f(x)=x^2, \forall x\in[M,M]$: uniformly continuous.
  (M fixed, >0)

Let's prove the third bullet point:

Let $x,y\in[-M,M]$. Then,

$$|f(x)-f(y)| = |x^2-y^2| = |(x-y)\cdot(x+y)| =$$
$$= |x-y|\cdot \underbrace{|x+y|}_{\leq |x|+|y|\leq 2M} \leq 2M\cdot|x-y|.$$

Let $\varepsilon>0$. If $x,y\in[-M,M]$, with $|x-y|<\frac{\varepsilon}{2M}$, then, by the above,

$$|f(x)-f(y)| < 2M\cdot\frac{\varepsilon}{2M} = \varepsilon.$$

Since $\delta(\varepsilon)=\frac{\varepsilon}{2M}$ only depends on ε, f is uniformly continuous.

We generalise this in <u>both</u> the Propositions that follow:

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→ **Def:** (Lipschitz continuity):

Let $f: A \subseteq \mathbb{R} \to \mathbb{R}$. We say that f is Lipschitz continuous if there exists $M>0$ such that

$$|f(x)-f(y)| \leq M \cdot |x-y|, \quad \forall x,y \in A.$$

→ **Prop:** Every Lipschitz continuous function is uniformly continuous.

Proof: Let $f: A \subseteq \mathbb{R} \to \mathbb{R}$ be Lipschitz continuous.

Then, there exists $M>0$ such that:

$$|f(x)-f(y)| \leq M \cdot |x-y|, \quad \forall x,y \in A.$$

Let $\varepsilon>0$. We choose $\delta = \frac{\varepsilon}{M}$ (depends only on ε).

If $x,y \in A$, and $|x-y|<\delta$, then

$$|f(x)-f(y)| \leq M \cdot |x-y| < M \cdot \delta = M \cdot \frac{\varepsilon}{M} = \varepsilon,$$

i.e. $|f(x)-f(y)|<\varepsilon$.

Since $\varepsilon>0$ was arbitrary, f is uniformly continuous. ∎

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→ works for open intervals (as long as f' bounded)

Prop: Let $f: I \to \mathbb{R}$, f continuous on I, differentiable in the interior of I. (I an interval)

If f' is bounded, then f is Lipschitz continuous (and thus uniformly continuous).

Proof: By our assumptions, there exists $M>0$ s.t.

$$|f'(y)| \leq M, \ \forall y \text{ in the interior of } I. \quad (*)$$

(WLOG) Let $x, y \in I$, with $x<y$. Since f is continuous on $[x,y]$ and differentiable on (x,y), we can apply the mean value theorem on $[x,y]$:

$$f(y) - f(x) = f'(\zeta) \cdot (y-x),$$

thus $|f(y) - f(x)| = |f'(\zeta)| \cdot |y-x|$, for some ζ between x and y, thus in the interior of I.

By $(*)$, $|f(y) - f(x)| \leq M \cdot |y-x|$.

Since $x, y \in I$ were arbitrary, f is Lipschitz continuous.

Therefore, f is uniformly continuous. ■

→ **ex:** $f(x) = \sin x$, $\forall x \in \mathbb{R}$: $|f'(x)| = |\cos x| \leq 1 \ \forall x \in \mathbb{R}$, thus f Lipschitz continuous (an interval), thus f uniformly continuous.

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Thm: Characterisation of uniform continuity via sequences:

Let $f: A \subseteq \mathbb{R} \to \mathbb{R}$.

f is uniformly continuous

$\iff$

for any pair of sequences $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}}$ in A,

with $x_n - y_n \longrightarrow 0$,

we have $f(x_n) - f(y_n) \longrightarrow 0$.

Proof: ($\Longrightarrow$) Let $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}}$ be in A, with $x_n - y_n \longrightarrow 0$. We want to show that $f(x_n) - f(y_n) \longrightarrow 0$:

Let $\varepsilon > 0$. Since f is uniformly continuous, we have that: for this $\varepsilon > 0$, there exists $\delta (= \delta(\varepsilon)) > 0$, such that: if $|x_n - y_n| < \delta$, then $|f(x_n) - f(y_n)| < \varepsilon$. (*)

Since $x_n - y_n \longrightarrow 0$, there exists $n_0 \in \mathbb{N}$ s.t.:

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$\forall n \geq n_0, \quad |x_n - y_n| < \delta.$

By (*): $\forall n \geq n_0, \quad |f(x_n) - f(y_n)| < \varepsilon.$

Since $\varepsilon > 0$ was arbitrary, $f(x_n) - f(y_n) \longrightarrow 0.$

($\Longleftarrow$) Suppose that f is not uniformly continuous.

We will construct sequences $(x_n)_{n\in\mathbb{N}}$, $(y_n)_{n\in\mathbb{N}}$ in A, with $x_n - y_n \longrightarrow 0$, but $f(x_n) - f(y_n) \not\longrightarrow 0$.

even better: $|f(x_n) - f(y_n)| \geq \varepsilon$, for some fixed $\varepsilon > 0$

Indeed: Since f doesn't satisfy the definition of uniform continuity:

there exists $\varepsilon > 0$ s.t. : $\forall \delta > 0,$

there exist $x_\delta, y_\delta \in A$, with $|x_\delta - y_\delta| < \delta,$

yet $|f(x_\delta) - f(y_\delta)| \geq \varepsilon.$ (*')

We apply (*') for $\delta = 1, \frac{1}{2}, \frac{1}{3}, \ldots$:

for $\delta = 1$: $\exists x_1, y_1 \in A$, with $|x_1 - y_1| < 1$, but $|f(x_1) - f(y_1)| \geq \varepsilon.$

for $\delta = 1/2$: $\exists x_2, y_2 \in A$, with $|x_2 - y_2| < \frac{1}{2}$, but $|f(x_2) - f(y_2)| \geq \varepsilon.$

$\vdots$

for $\delta = 1/n$: $\exists x_n, y_n \in A$, with $|x_n - y_n| < \frac{1}{n}$, but $|f(x_n) - f(y_n)| \geq \varepsilon$

$\vdots$

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Since $|x_n - y_n| < \frac{1}{n} \quad \forall n \in \mathbb{N}$,

we have $x_n - y_n \longrightarrow 0$ (by the sandwich lemma).

Thus, by our assumption, we should have

$$f(x_n) - f(y_n) \longrightarrow 0.$$

However, $|f(x_n) - f(y_n)| \geq \underbrace{\varepsilon}_{\text{fixed}} > 0, \quad \forall n \in \mathbb{N},$

thus $f(x_n) - f(y_n) \not\longrightarrow 0.$

This is a contradiction. Therefore, f is uniformly continuous.

■

→ This characterisation of uniform continuity can prove particularly useful when we want to show that a function f is **not** uniformly continuous.

→ Example: Show that $f: (0,1) \longrightarrow \mathbb{R}$, with $f(x) = \frac{1}{x}, \forall x \in (0,1)$, is not uniformly continuous.

! METHOD !

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Solution: We will find $(x_n)_{n\in\mathbb{N}}$, $(y_n)_{n\in\mathbb{N}}$ in $(0,1)$, with $x_n - y_n \longrightarrow 0$, but $f(x_n) - f(y_n) \not\longrightarrow 0$.

Indeed: Let $x_n = \frac{1}{n}$ $\forall n\in\mathbb{N}$,

and $y_n = \frac{1}{2n}$ $\forall n\in\mathbb{N}$.

The sequences $(x_n)_{n\in\mathbb{N}}$, $(y_n)_{n\in\mathbb{N}}$ are both in $(0,1)$.

And: $x_n - y_n = \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n} \longrightarrow 0$.

But: $f(x_n) - f(y_n) = \frac{1}{x_n} - \frac{1}{y_n} = n - 2n = -n \longrightarrow -\infty \neq 0$

Thus, f is not uniformly continuous.

→ You can try a similar trick to show that

$f(x) = x^2$ $\forall x\in\mathbb{R}$ and $g(x) = \cos(x^2)$ $\forall x\in\mathbb{R}$

are not uniformly continuous.

(Exercise).

⚠ A bounded and continuous function is not necessarily uniformly continuous (ex.: $g(x) = \cos(x^2)$, $\forall x\in\mathbb{R}$).

- $f: I \longrightarrow \mathbb{R}$ (I open interval) continuous is not necessarily uniformly continuous. (ex.: $f(x) = \frac{1}{x}$, $\forall x\in(0,1)$).

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→ works only for closed intervals.

Thm: Every continuous function $f : [a,b] \to \mathbb{R}$ is uniformly continuous.

Proof: Suppose that f is not uniformly continuous.

Then, there exists $\varepsilon > 0$ and $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}}$ in $[a,b]$, such that $x_n - y_n \to 0$, but $|f(x_n) - f(y_n)| \geq \varepsilon, \forall n \in \mathbb{N}$.

(follows from proof of characterisation via sequences. Stronger than just $f(x_n) - f(y_n) \not\to 0$!)

We will show that this is a contradiction.

Indeed:

- $(x_n)_{n\in\mathbb{N}}$ is bounded, thus, by Bolzano-Weierstrass, it has a convergent subsequence $(x_{k_n})_{n\in\mathbb{N}}$.
- That is, $x_{k_n} \to x_0 \in \mathbb{R}$, for this $(x_{k_n})_{n\in\mathbb{N}}$.

  Since $a \leq x_{k_n} \leq b \quad \forall n \in \mathbb{N}$,

  we have $x_0 \in [a,b]$. Therefore, f continuous at x_0 $\Longrightarrow f(x_{k_n}) \to f(x_0)$.
- Since $x_n - y_n \to 0$, we have $x_{k_n} - y_{k_n} \to 0$ as well.

  And since in addition $x_{k_n} \to x_0$, we have $y_{k_n} \to x_0$.

  $\left(y_{k_n} = x_{k_n} - (x_{k_n} - y_{k_n}) \to x_0 - x_0 = 0. \right)$

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Since f is continuous at x_0, it follows that

$$f(y_{k_n}) \longrightarrow f(x_0).$$

Therefore:

$$\left.\begin{array}{l} f(x_{k_n}) \longrightarrow f(x_0) \\ f(y_{k_n}) \longrightarrow f(x_0) \end{array}\right\} \longrightarrow f(x_{k_n}) - f(y_{k_n}) \longrightarrow f(x_0) - f(x_0) = 0.$$

However, this is a contradiction: we picked $(x_n)_{n\in\mathbb{N}}, (y_n)_{n\in\mathbb{N}}$ such that $|f(x_{k_n}) - f(y_{k_n})| \geq \varepsilon > 0$, $\forall n \in \mathbb{N}$, thus

$$f(x_{k_n}) - f(y_{k_n}) \not\longrightarrow 0 \quad (\varepsilon \text{ fixed}).$$

Therefore, f is uniformly continuous.

■

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Unitary matrices

As we have seen in class, an $n \times n$ unitary matrix $\boldsymbol{U}$ is a matrix obeying

$$\boldsymbol{U}^*\boldsymbol{U} = \boldsymbol{I}. \tag{1}$$

Expressed differently, a unitary matrix is a matrix whose inverse is its adjoint. If the columns of $\boldsymbol{U}$ are the vectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n$, the (i,j)th entry of $\boldsymbol{U}^*\boldsymbol{U}$ is equal to $\langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle$ and, therefore, another way to write (1) is

$$\langle \boldsymbol{u}_i, \boldsymbol{u}_j \rangle = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

This says that the family $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ is an orthonormal basis of $\mathbb{C}^n$. In short, the columns of a unitary matrix form an orthobasis.

Now recall that when we write $\boldsymbol{U}\boldsymbol{x} = \boldsymbol{b}$, or equivalently

$$x_1\boldsymbol{u}_1 + \ldots + x_n\boldsymbol{u}_n = \boldsymbol{b}, \tag{2}$$

we can think of $\boldsymbol{x}$ as the coordinates of the vector $\boldsymbol{b}$ in the basis $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$. These coordinates are given by

$$\boldsymbol{x} = \boldsymbol{U}^{-1}\boldsymbol{b} = \boldsymbol{U}^*\boldsymbol{b} \quad \Leftrightarrow \quad x_i = \langle \boldsymbol{u}_i, \boldsymbol{b} \rangle, \ i = 1, \ldots, n.$$

Hence, we can rewrite (2) as

$$\langle \boldsymbol{u}_1, \boldsymbol{b} \rangle \boldsymbol{u}_1 + \ldots + \langle \boldsymbol{u}_n, \boldsymbol{b} \rangle \boldsymbol{u}_n = \boldsymbol{b}.$$

Observe that we can rewrite the left-hand side as $(\boldsymbol{u}_1\boldsymbol{u}_1^* + \ldots + \boldsymbol{u}_n\boldsymbol{u}_n^*)\boldsymbol{b}$ so that

$$(\boldsymbol{u}_1\boldsymbol{u}_1^* + \ldots + \boldsymbol{u}_n\boldsymbol{u}_n^*)\boldsymbol{b} = \boldsymbol{b}.$$

Since this holds *for all* $\boldsymbol{b} \in \mathbb{C}^n$, this means nothing else than

$$\boldsymbol{u}_1\boldsymbol{u}_1^* + \ldots + \boldsymbol{u}_n\boldsymbol{u}_n^* = \boldsymbol{I}. \tag{3}$$

In conclusion, whenever $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_n\}$ is an orthobasis, (3) always holds.

Suppose now that you give me $m \leq n$ orthonormal vectors $\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m$. Then the orthonormal projection onto the span of these vectors is given by

$$\boldsymbol{P} = \boldsymbol{u}_1\boldsymbol{u}_1^* + \ldots + \boldsymbol{u}_m\boldsymbol{u}_m^*.$$

When $m < n$, this cannot be the identity matrix because the column space of $\boldsymbol{P}$ is $\operatorname{span}(\boldsymbol{u}_1, \ldots, \boldsymbol{u}_m)$, which is of dimension $m < n$.

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MTH631 Mid Term Past Paper By a Student

(VISIT VURANK FOR MORE)

Spring 2018

MCQS:

MCQS were randomly asked 4, 5 were from quizzes.

Questions:

1) Definition of closure of set, closed ball. (5 marks)

2) Cauchy convergence criteria. (5 marks)

3) One question is about proof of some formula? (10 marks)

(THE END)

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MTH631 Mid Term Past Paper By a Student

(VISIT VURANK FOR MORE)

Spring 2018

MCQS:

Total 21 MCQS.

Questions:

1) Prove Schwartz's inequality?

2) One long is from Foureir Series. (5 marks)

3) One is from point of convergence. (5 marks)

4) Find first four non-zero terms of McLaurin series of sinxlog (1+x)? (10 marks)

5) An Example of weierstrass M-test (Read this topic carefully).

(THE END)

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